can be signed, the contents of the JWT is first hashed using a cryptographic hash function and then the hash value is signed. JWS provides variants for different hash functions when using the same underlying signature algorithm. All the allowed hash functions provide adequate security, but SHA-512 is the most secure and may be slightly faster than the other choices on 64-bit systems. The exception to this rule is when using ECDSA signatures, because JWS specifies elliptic curves to use along with each hash function; the curve used with SHA-512 has a significant performance penalty compared with the curve used for SHA-256.

Table 7.2 JWS signature algorithms

JWS Algorithm	Hash function	Signature algorithm
RS256	SHA-256	
RS384	SHA-384	RSA with PKCS#1 v1.5 padding
RS512	SHA-512	
PS256	SHA-256	
PS384	SHA-384	RSA with PSS padding
PS512	SHA-512	
ES256	SHA-256	ECDSA with the NIST P-256 curve
ES384	SHA-384	ECDSA with the NIST P-384 curve
ES512	SHA-512	ECDSA with the NIST P-521 curve
EdDSA	SHA-512 / SHAKE256	EdDSA with either the Ed25519 or Ed448 curves

Of these choices, the best is EdDSA, based on the Edwards Curve Digital Signature Algorithm (https://tools.ietf.org/html/rfc8037). EdDSA signatures are fast to produce and verify, produce compact signatures, and